

1. Oscillation Verification

Objective

To verify that the designed 5-stage CMOS Ring Oscillator generates sustained oscillations after power-up.

Measurement

- Performed transient simulation using:
`.tran 0 100n startup`
- Applied an initial condition to start oscillation.

Observation

A stable periodic square waveform was observed at the output node after a short startup interval.

Reason

The odd number of inverter stages introduces a total phase shift of 180° , while the propagation delay contributes an additional 180° phase shift, satisfying the Barkhausen criterion for oscillation.

Conclusion

The ring oscillator successfully starts and maintains continuous oscillation.

2. Frequency Measurement

Objective

To determine the oscillation frequency of the ring oscillator.

Measurement

```
.meas tran tperiod TRIG V(osc) VAL=0.9 RISE=50 TARG V(osc) VAL=0.9 RISE=60  
.meas tran frequency PARAM='10/tperiod'
```

Observation

Measured Frequency:

- 5-stage: **1.64 GHz**
- 9-stage: **910.8 MHz**

Reason

The oscillation frequency is inversely proportional to the total propagation delay around the loop.

$$f = \frac{1}{2Nt_{pd}}$$

Increasing the number of stages increases the total delay, thereby reducing the oscillation frequency.

Conclusion

Measured frequencies closely match theoretical predictions.

3. Propagation Delay

Objective

To determine the propagation delay of a CMOS inverter stage.

Measurement

```
.meas tran tphl ...
```

```
.meas tran tplt ...
```

```
.meas tran tpd PARAM='(tphl+tplt)/2'
```

Observation

Average propagation delay

≈ 57 ps

Reason

Propagation delay depends on transistor drive current and load capacitance.

Larger drive current reduces delay.

Higher load capacitance increases delay.

Conclusion

The measured delay agrees well with theoretical calculations.

4. Startup Analysis

Objective

To determine the startup characteristics of the oscillator.

Measurement

Transient analysis

.tran 0 100n startup

Observation

Oscillation starts after a brief transient and reaches a stable amplitude.

Reason

The startup option removes the DC operating point, allowing numerical noise or the specified initial condition to initiate oscillation.

Conclusion

The oscillator starts reliably under the specified operating conditions.

5. Frequency vs Temperature

Objective

To analyze the effect of temperature on oscillation frequency.

Measurement

.step temp list 0 25 50 75 100

Measured frequency at each temperature.

Observation

Frequency decreases with increasing temperature.

Reason

Higher temperature reduces carrier mobility, decreasing transistor drive current. This increases propagation delay, which in turn lowers the oscillation frequency.

Conclusion

The oscillator exhibits the expected thermal behavior of CMOS circuits.

6. Delay vs Temperature

Objective

To evaluate the effect of temperature on propagation delay.

Measurement

Measured average propagation delay for each temperature.

Observation

Propagation delay increases as temperature increases.

Reason

Reduced carrier mobility slows transistor switching, increasing the charging and discharging time of node capacitances.

Conclusion

Propagation delay shows a positive temperature coefficient.

7. Power vs Temperature

Objective

To evaluate power consumption over temperature.

Measurement

```
.meas tran lavg AVG I(V1)
```

```
.meas tran Pavg PARAM='1.8*abs(lavg)'
```

Observation

Average power decreases with increasing temperature.

Reason

Although leakage current increases with temperature, the oscillation frequency decreases significantly. The reduction in dynamic switching power dominates, leading to lower average power consumption.

Conclusion

Dynamic power dominates total power consumption in the simulated operating range.

8. Average Current Measurement

Objective

To measure the average current drawn from the supply.

Measurement

```
.meas tran lavg AVG I(V1)
```

Observation

Average supply current decreases with increasing temperature.

Reason

Lower oscillation frequency results in fewer switching events per second, reducing average current.

Conclusion

Current consumption follows the trend of oscillation frequency.

9. Frequency vs Supply Voltage

Objective

To study the effect of supply voltage on oscillation frequency.

Measurement

```
.param VDD=1.8
```

```
.step param VDD list 1.2 1.4 1.6 1.8 2.0
```

Observation

Frequency increases as supply voltage increases.

Reason

Higher supply voltage increases transistor overdrive voltage, producing higher drive current and reducing propagation delay.

Conclusion

Supply voltage has a significant influence on oscillator speed.

10. Load Capacitance Sweep

Objective

To analyze the effect of capacitive loading.

Measurement

```
.step param CL list 10f 20f 50f 100f 200f
```

Observation

Frequency decreases with increasing load capacitance.

Reason

A larger load requires more time to charge and discharge, increasing propagation delay.

Conclusion

The oscillation frequency is inversely related to load capacitance.

11. Transistor Sizing Optimization**Objective**

To determine the effect of transistor dimensions on performance.

Measurement

Repeated simulations for different NMOS and PMOS widths while maintaining constant channel length.

Observation

Increasing transistor width increases oscillation frequency but also increases power consumption.

Reason

Larger transistors provide higher drive current, reducing delay, but introduce greater gate and diffusion capacitance and consume more dynamic power.

Conclusion

A trade-off exists between speed, power, and area.

12. Output Swing**Objective**

To verify logic voltage levels.

Measurement

Measured maximum and minimum output voltages.

Observation

Output swings from approximately 0 V to VDD.

Reason

Complementary NMOS and PMOS transistors provide full rail-to-rail operation.

Conclusion

The oscillator produces valid CMOS logic levels.

13. Duty Cycle

Objective

To determine waveform symmetry.

Measurement

Measured high and low pulse widths.

Observation

Duty cycle is close to 50%.

Reason

Balanced NMOS and PMOS sizing produces nearly equal rise and fall delays.

Conclusion

Balanced duty cycle confirms proper inverter sizing.

14. Energy per Cycle

Objective

To evaluate the energy consumed during one oscillation cycle.

Measurement

Calculated using

$$E = \frac{P}{f}$$

Observation

Energy per cycle remained approximately constant over the measured temperature range.

Reason

Both power and frequency decrease proportionally with increasing temperature.

Conclusion

Energy per oscillation cycle provides an effective measure of oscillator efficiency.

15. Power-Delay Product (PDP)

Objective

To evaluate energy efficiency.

Measurement

Calculated as

$$PDP = P \times t_{pd}$$

Observation

The calculated PDP indicates the energy required for one switching event.

Reason

PDP combines power consumption and propagation delay into a single performance metric.

Conclusion

A lower PDP indicates a more energy-efficient design and is widely used for comparing CMOS circuits.

16. Theoretical Validation

Objective

To compare measured results with theoretical calculations.

Measurement

Theoretical frequency:

$$f = \frac{1}{2Nt_{pd}}$$

Observation

The measured frequency closely matched the theoretical value, with only a small deviation.

Reason

The difference arises from transistor nonlinearity, parasitic capacitances, unequal rise/fall delays, and numerical simulation effects.

Conclusion

The close agreement validates the accuracy of the implemented design and simulation methodology.